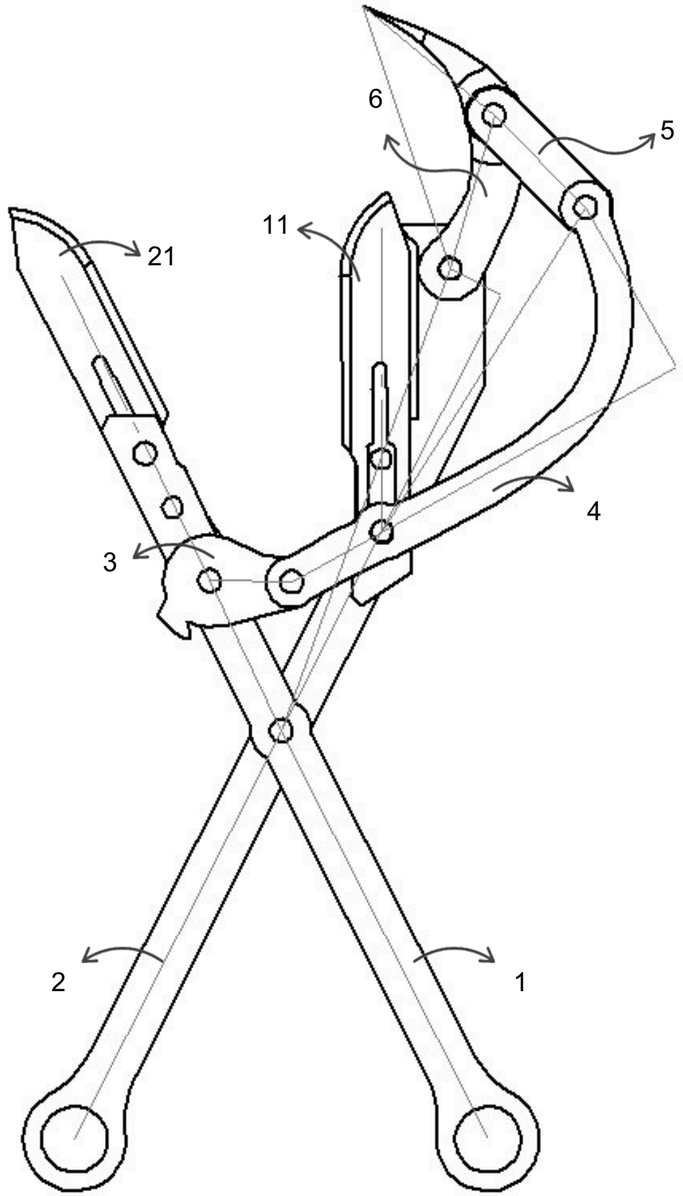
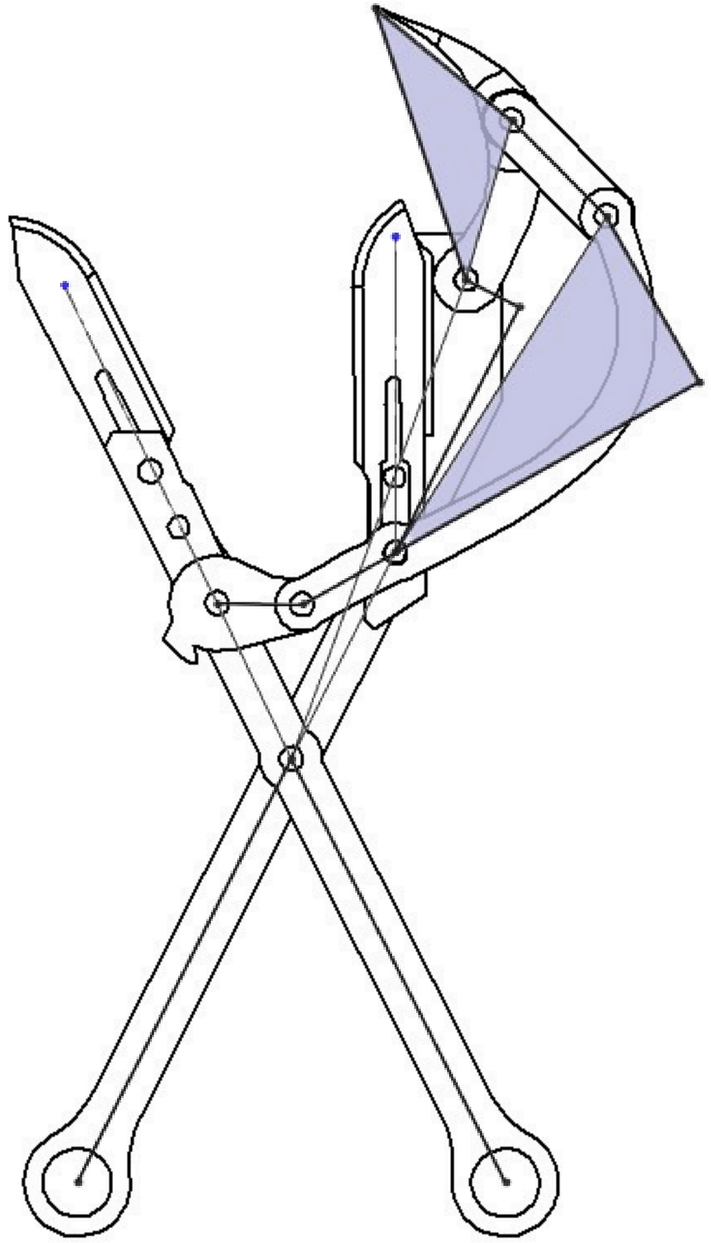
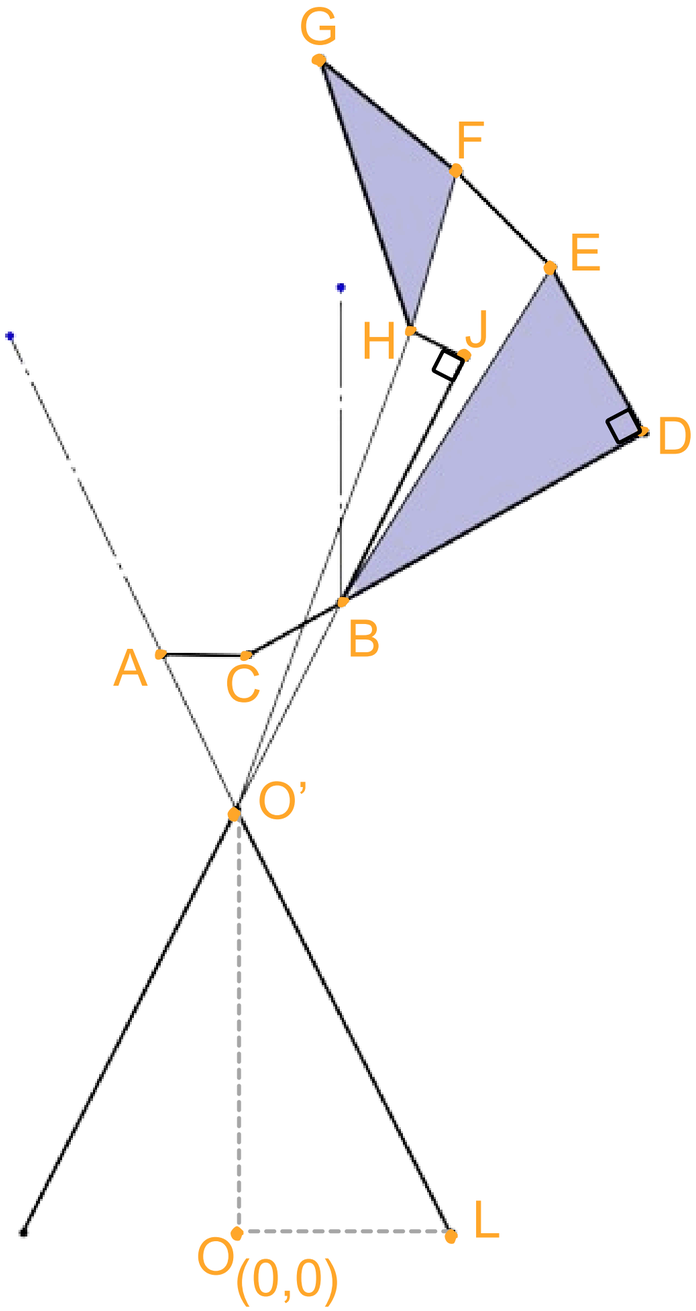
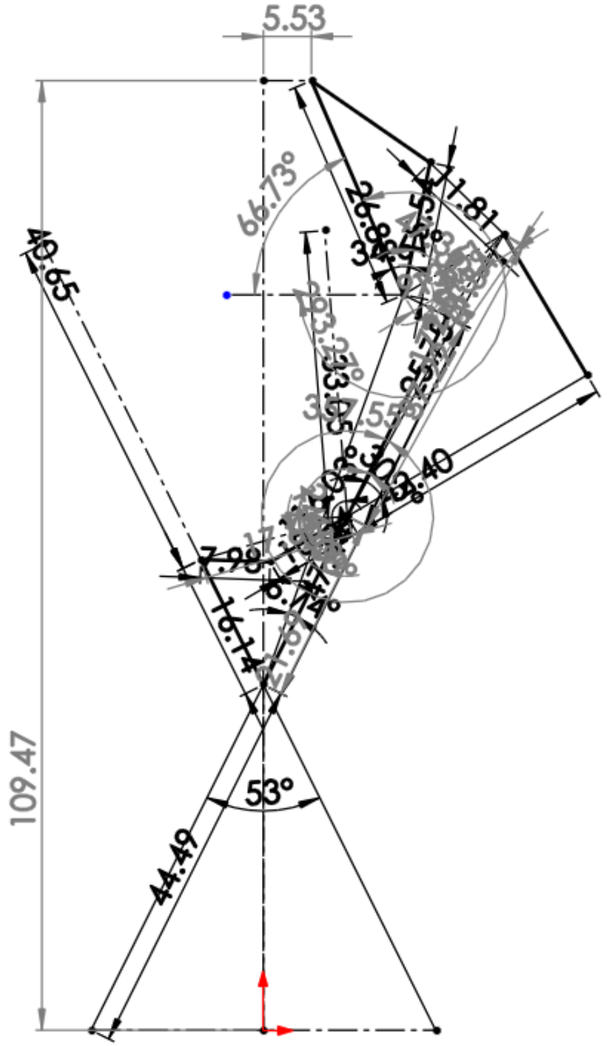

6
5
11
21
4
3
2
1




G
F
E
H
J
D
A
C
B
O’
L
O
(0,0)

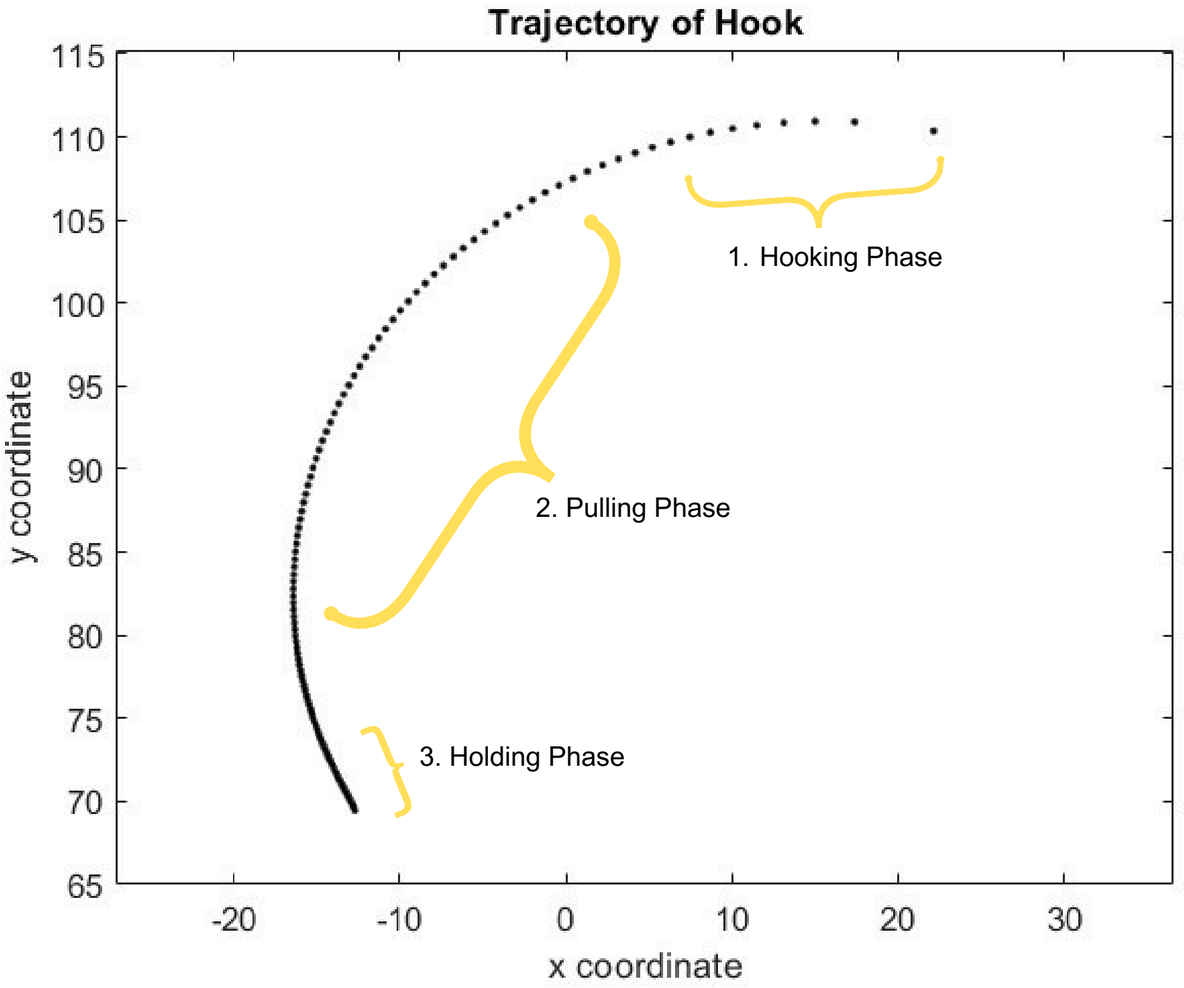


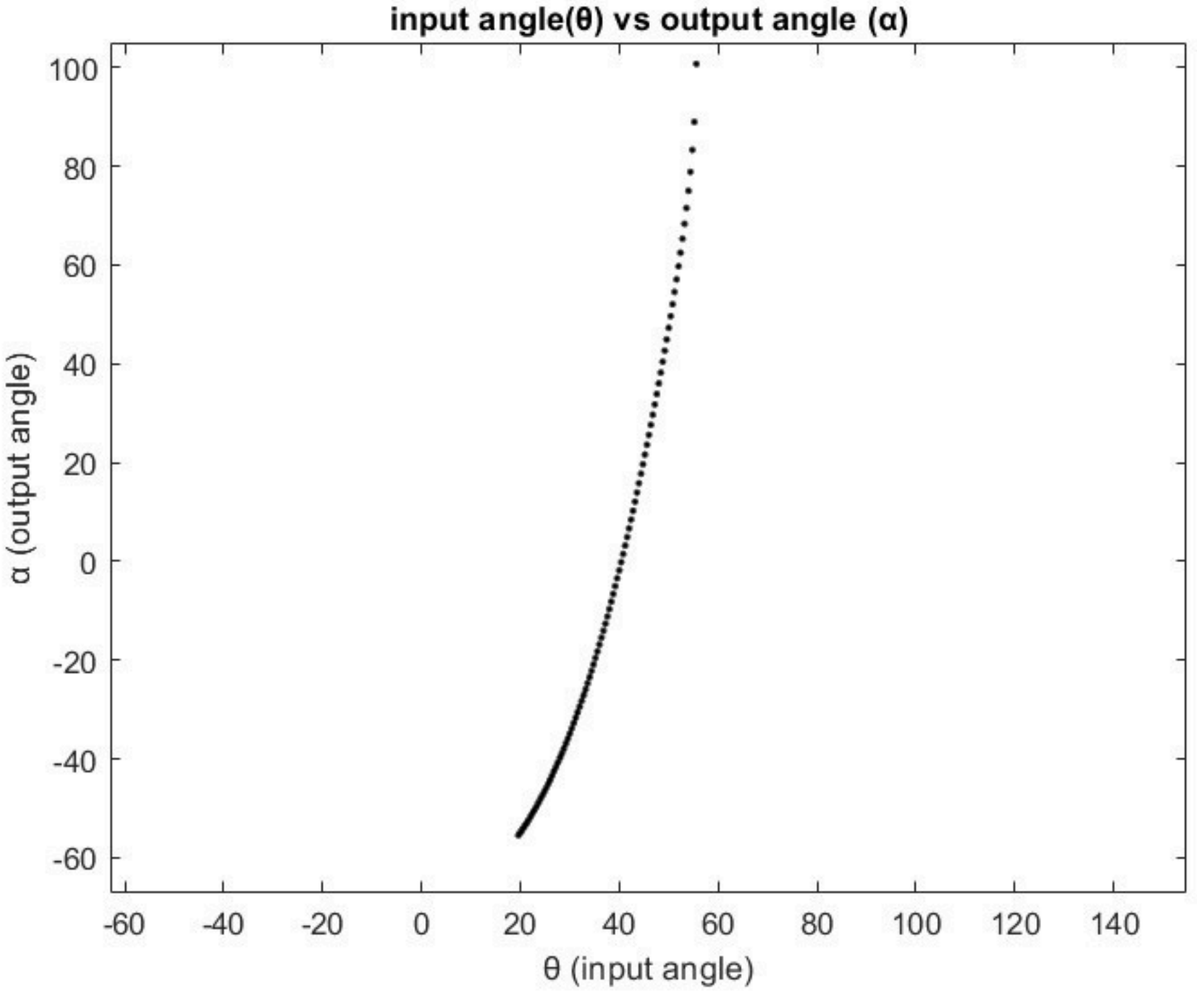
For Input angle θ = 53°	output angle α	Hook end x coordinate	Hook end y coordinate
Result by kinematically constraint sketch in SolidWorks	66.7358	5.5334	109.4732
Result by the derived governing equation plotted in MATLAB	66.7359	5.5336	109.4734

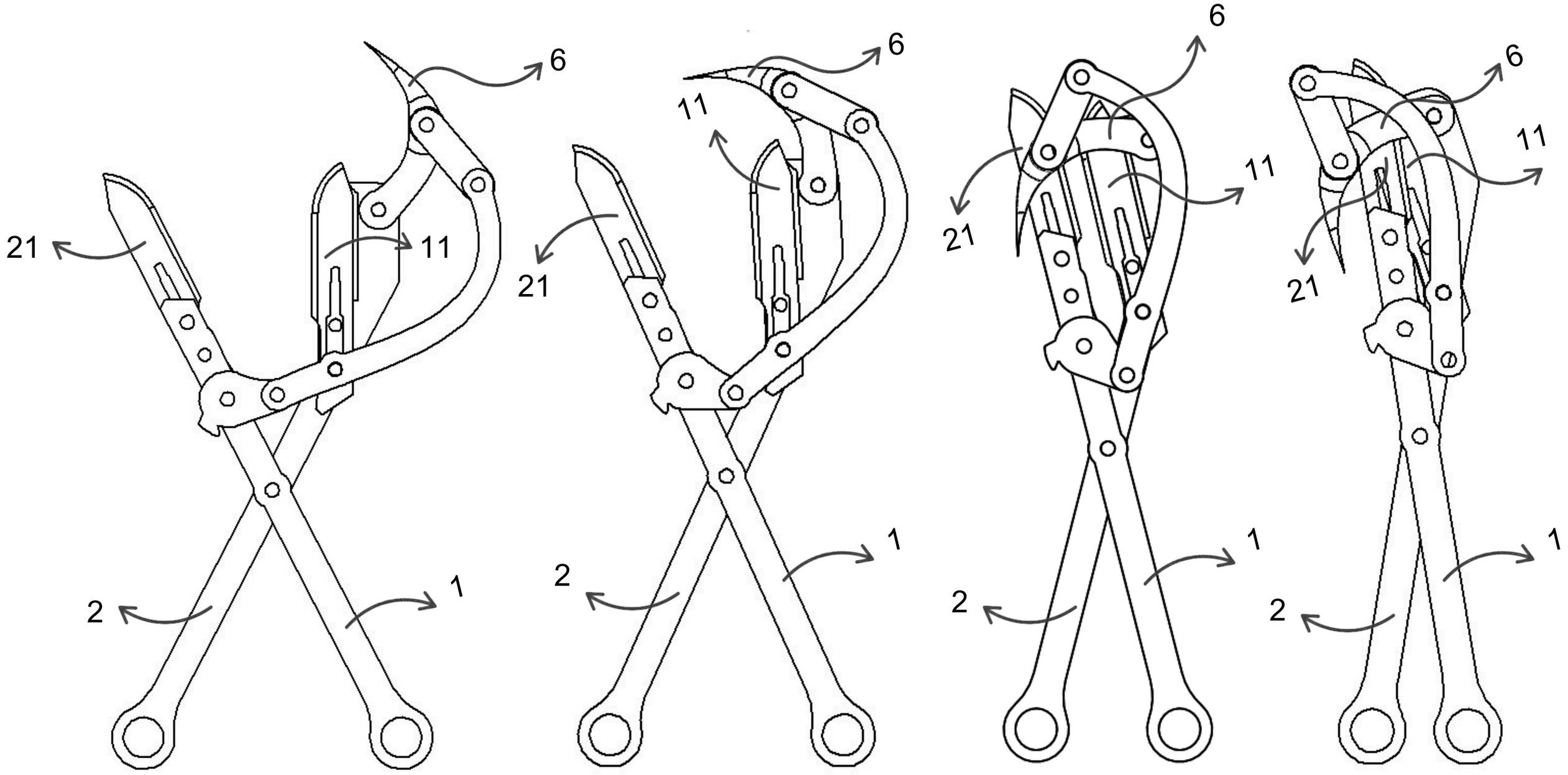
For Input angle θ = 38°	output angle α	Hook end x coordinate	Hook end y coordinate
Result by kinematically constraint sketch in SolidWorks	-9.6729	-16.3539	84.2363
Result by the derived governing equation plotted in MATLAB	-9.6731	-16.3540	84.2365

For Input angle θ = 23°	output angle α	Hook end x coordinate	Hook end y coordinate
Result by kinematically constraint sketch in SolidWorks	-50.0639	-13.2914	70.5978
Result by the derived governing equation plotted in MATLAB	-50.0640	-13.2915	70.5980




Trajectory of Hook
y coordinate
x coordinate
115
110
105
100
95
90
85
80
75
70
65
-20
-10
0
10
20
30
1. Hooking Phase
2. Pulling Phase
3. Holding Phase


input angle(θ) vs output angle (α)
α (output angle)
θ (input angle)
100
80
60
40
20
0
-20
-40
-60
-60
-40
-20
0
20
40
60
80
100
120
140


6
21
11
2
1
6
11
21
1
2
6
11
21
1
2
6
11
21
1
2

$$x = \frac{l_2 \sin(\theta/2 - \lambda)}{\cos\lambda} - d\cos\alpha \quad \& \quad y = l_1 \cos(\theta/2) + \frac{l_2 \cos(\theta/2 - \lambda)}{\cos\lambda} + d\sin\alpha$$

where :-

$$\alpha = \pi - \psi_2 + \theta/2 + \cos^{-1}\left(\frac{(l_2 \tan\lambda)^2 + b_3^2 - b_2^2}{2 * l_2 \tan\lambda * b_3}\right) - \cos^{-1}\left(\frac{a_6^2 + b_3^2 - a_7^2}{2 a_6 b_3}\right)$$

$$b_3 = \sqrt{(l_2 \tan\lambda)^2 + b_2^2 - 2 * l_2 \tan\lambda * b_2 \cos\gamma_5}$$

$$b_2 = \sqrt{a_5^2 + l_3^2 - 2 a_5 l_3 \cos\Phi_4}$$

$$\gamma_5 = \pi/2 + \sin^{-1}(a_5 \sin\Phi_4 / b_2)$$

$$\Phi_4 = \sin^{-1}(a_1 \sin\theta / b_1) - \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2 a_4 b_1}\right) - \psi_1$$

$$b_1 = \sqrt{a_1^2 + a_2^2 - 2 a_1 a_2 \cos\theta}$$

Tomato Cutting Scissor Mechanism

Shobit Verma

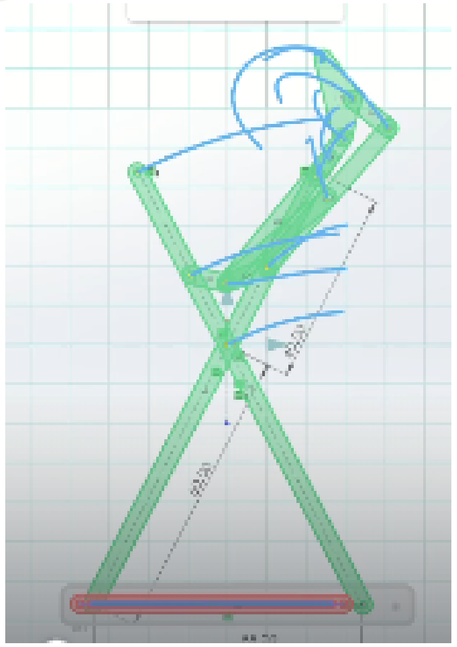
DESIGN REPORT

Brief Background

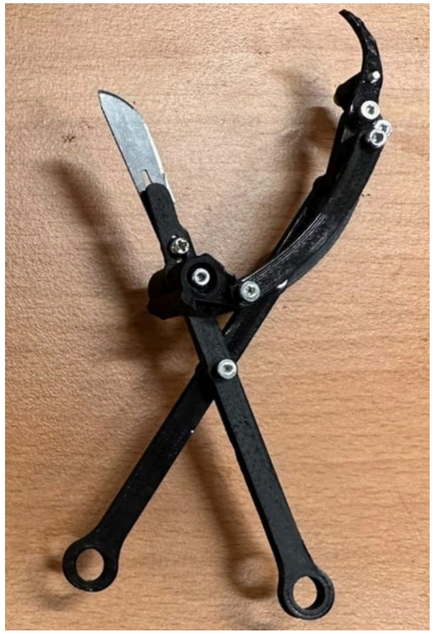
During the autonomous cutting of tomatoes we often face the problem with the cutting end effector. These systems usually rely on point cloud visual data to identify the cutting point. These points often have discrepancies, leading to inaccuracies in the exact position of the cutting point. This can result in incomplete cuts or damage to the fruit itself.

Thus we need to have a hook mechanism which pulls out the pedicel from a zone of possibilities of occurrence as per the detected cutting point. Existing tools and automated systems also often lack the capability to securely pull and stabilize the pedicel during cutting, which is crucial for ensuring a clean and precise cut.

Previous models



initial concept - 10 bar



Proto 3- 8 bar



Proto 4- 8 bar



Proto 7- 8 bar

(Field tested version)

Problem with proto 7 in field test

Theoretical was working good but in practical -

- hook does not pull well
- too complex cannot exert force much
- singularities occur during operation





Newer versions -



Proto 7- 8 bar

(Field tested version)



Proto 8- 6 bar



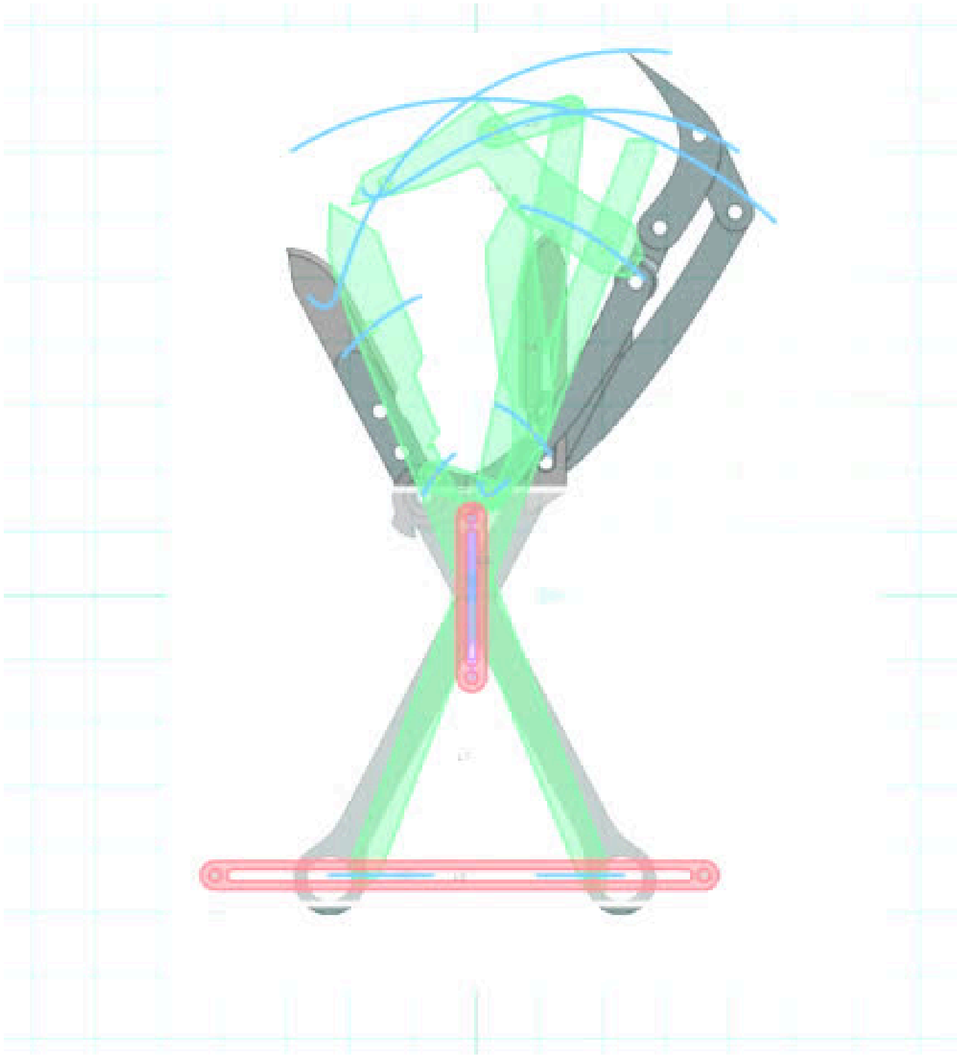
Proto 9- 6 bar



Proto 10- 6 bar

Simulation

- Simpler mechanism, good amount of force transmit
- No singularities
- increased cutting space
- strengthened hooking
- Locking of the mechanism at the end to lock the stem in it.
- Mechanism is designed such that it cannot collapse fully kinematically to prevent unnecessary extra motion
- mechanism cannot extend fully as well kinematically to prevent overshoot
- Basically kinematically ranged mechanism



Working video



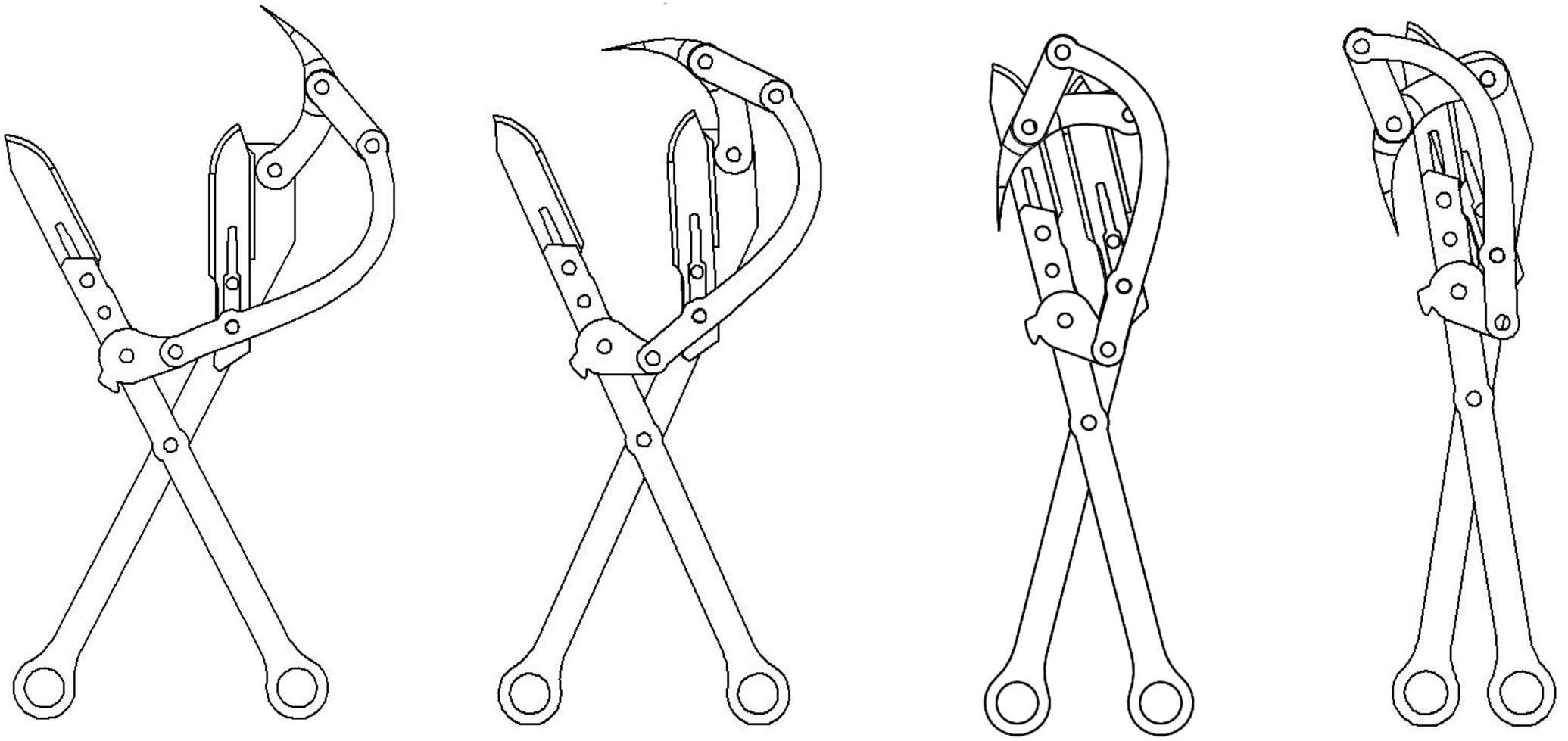
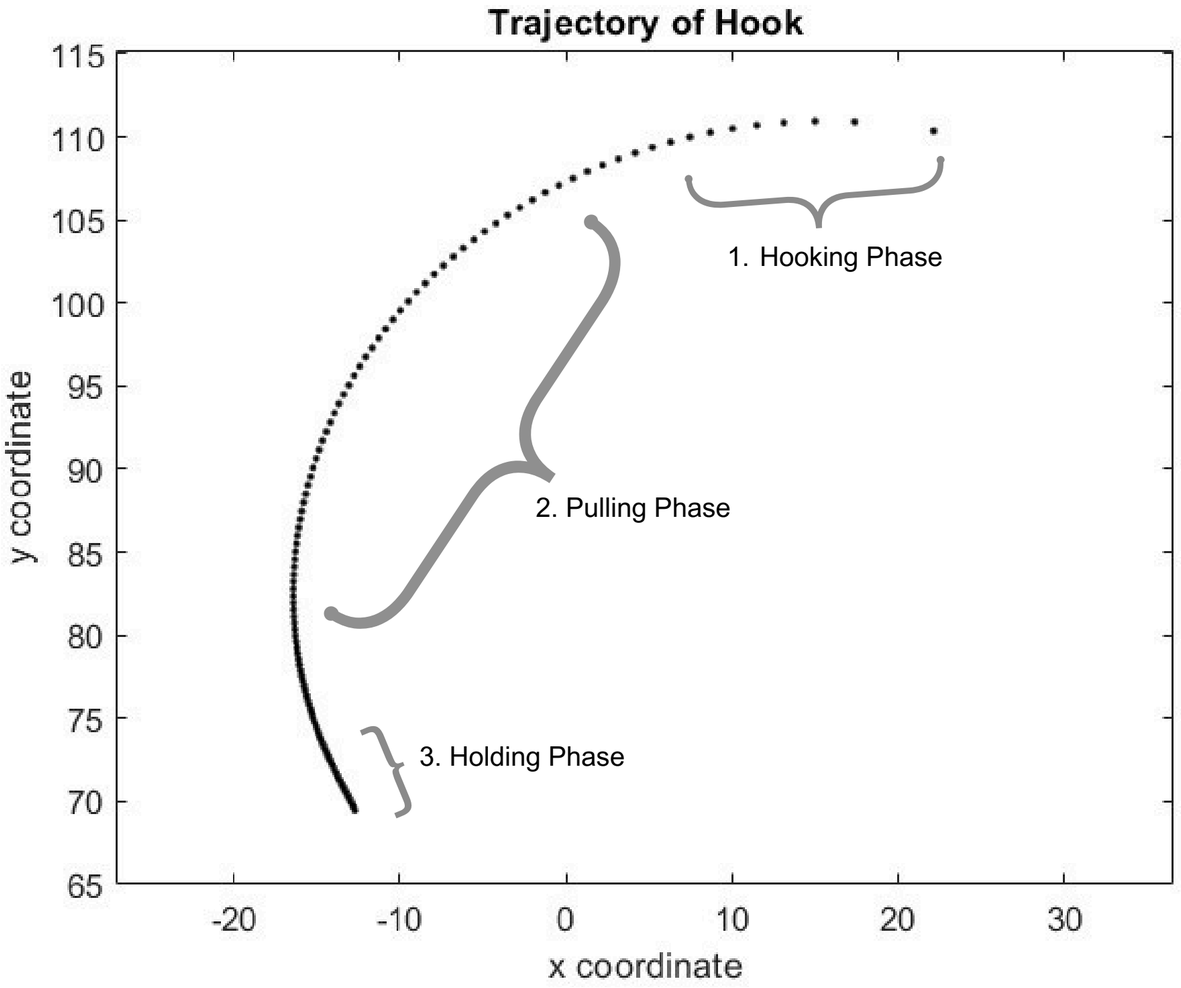
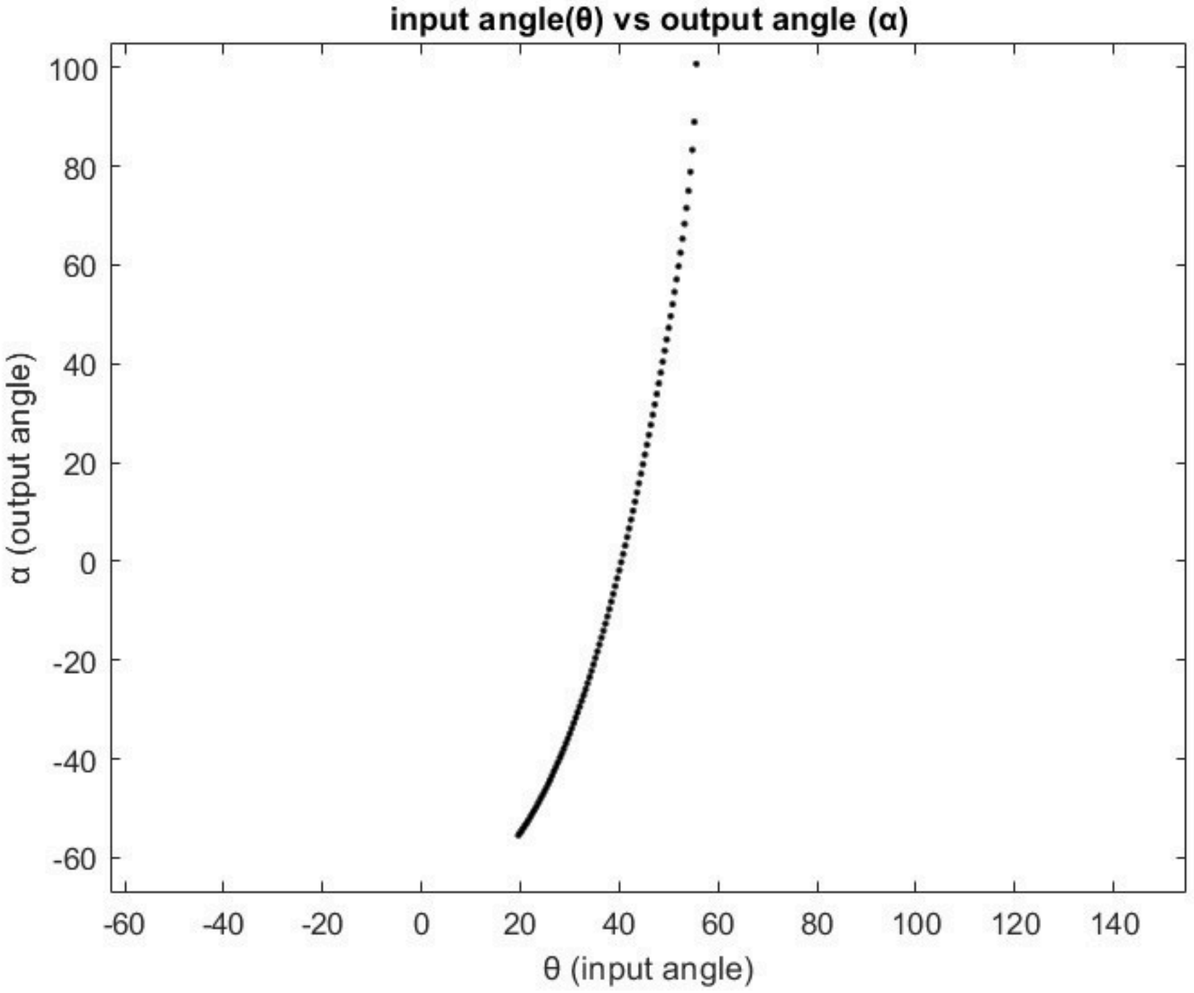


Fig: Kinematic motion of the cutter in phases with positions (a) initial position, (b) end of hooking phase, (c) end of pulling phase, (d) end of holding phase.


Trajectory of Hook
y coordinate
x coordinate
115
110
105
100
95
90
85
80
75
70
65
-20
-10
0
10
20
30
1. Hooking Phase
2. Pulling Phase
3. Holding Phase


input angle(θ) vs output angle (α)
α (output angle)
θ (input angle)
100
80
60
40
20
0
-20
-40
-60
-60
-40
-20
0
20
40
60
80
100
120
140